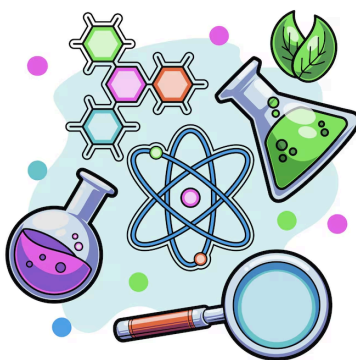




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational A • 2026



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. Which of the following occurs during anaphase of mitosis?
- A) Chromosomes decondense and nuclear envelopes reform.
 - B) DNA is replicated and the cell prepares for division.
 - C) Spindle fibers attach to centromeres and chromosomes align.
 - D) Chromosomes condense and spindle fibers begin to form.
 - E) Sister chromatids are pulled to opposite poles.
- B02. Phenotype refers to
- A) The number of gametes in an individual.
 - B) An individual's genetic composition.
 - C) The combination of alleles in an individual.
 - D) The observable expression of the genes in an individual.
 - E) The number of chromosomes in an individual.
- B03. _____ reactions release energy by breaking down complex molecules.
- A) Endergonic
 - B) Anabolic
 - C) Catabolic
 - D) Photosynthetic
 - E) Reductive
- B04. Which organ system is primarily responsible for transporting oxygen, nutrients, and waste throughout the body?
- A) Respiratory
 - B) Circulatory
 - C) Digestive
 - D) Nervous
 - E) Integumentary
- B05. Where does transcription of most prokaryotic genes primarily occur?
- A) Inside the nucleus.
 - B) On the rough endoplasmic reticulum.
 - C) Within the mitochondria.
 - D) In the cytoplasm.
 - E) On free ribosomes.
- B06. A population of beetles has two color morphs: green and brown. Birds preferentially prey on green beetles. Over several generations, the frequency of brown beetles increases. Which evolutionary mechanism best explains this change?
- A) Genetic drift
 - B) Gene flow
 - C) Natural selection
 - D) Mutation
 - E) Sexual selection
- B07. If helicase function is blocked during DNA replication, which process would be directly affected?
- A) Separation of parental DNA strands.
 - B) Rate of DNA synthesis.
 - C) Segregation of sister chromatids.
 - D) Transcription initiation at promoters.
 - E) Frequency of spontaneous mutations.
- B08. Which process in the nitrogen cycle converts atmospheric nitrogen (N_2) into ammonia (NH_3), making nitrogen available to living organisms?
- A) Nitrogen fixation
 - B) Denitrification
 - C) Ammonification
 - D) Nitrification
 - E) Assimilation

- B09. In November 2025, the Centers for Disease Control and Prevention announced a multistate outbreak of *Listeria* linked to
- A) peaches.
 - B) cookie dough.
 - C) pistachio ice cream.
 - D) Prepared pasta meals.
 - E) leafy greens.
- B10. During dehydration (condensation) synthesis, which of the following occurs?
- A) Monomers are broken apart by adding a molecule of water.
 - B) Water is used to cleave covalent bonds.
 - C) Covalent bonds are broken by hydrolytic enzymes.
 - D) Polymers are joined together by removing a molecule of water.
 - E) Monomers are joined together by removing a molecule of water.
- B11. In photosynthesis, if the chloroplasts are damaged, which immediate effect will occur?
- A) The plant will increase glucose production.
 - B) Cellular respiration rates will decrease.
 - C) The plant will be unable to convert light energy into chemical energy.
 - D) The plant will increase oxygen production.
 - E) The plant's water loss through transpiration will not change significantly.
- B12. A scientist inserts a human insulin gene into bacteria so the bacteria will produce insulin that can be harvested for medical use. This is an example of:
- A) Genetic engineering
 - B) Selective breeding
 - C) Mutation
 - D) Natural selection
 - E) Gene therapy
- B13. Two organisms belonging to the same Class would also belong to the same ____.
- A) Family
 - B) Order
 - C) Genus
 - D) Phylum
 - E) Order
- B14. In a study of human anatomy, scientists discovered a new, discrete structure that carries out specific functions (like filtering waste). To properly classify this structure, which term should they use, and why?
- A) Organ system — because it consists of multiple organs coordinating functions
 - B) Organ — because it is made of various tissues working as a functional unit
 - C) Gene — because it controls the development of tissues
 - D) Tissue — because it is a group of similar cells performing the same function
 - E) Molecule — because it is the smallest unit of biological function
- B15. Which of the following best describes commensalism in an ecological community?
- A) Both species benefit from the interaction.
 - B) One species benefits while the other is harmed.
 - C) Both species are harmed.
 - D) One species uses a resource without affecting its availability to others.
 - E) One species benefits while the other is unaffected.
- B16. Which of the following is the causative agent of whooping cough?
- A) *Mycoplasma pneumoniae*
 - B) *Bordetella pertussis*
 - C) *Haemophilus influenzae*
 - D) *Streptococcus pyogenes*
 - E) *Legionella pneumophila*

- B17. A patient suffers an injury that damages the tissue responsible for voluntary movement. Which tissue type was most likely injured?
- A) Cardiac tissue
 - B) Skeletal muscle tissue
 - C) Smooth muscle tissue
 - D) Nervous tissue
 - E) Connective tissue
- B18. Which of the following processes requires energy input from the cell to move substances across the plasma membrane?
- A) Osmosis of water.
 - B) Diffusion of oxygen.
 - C) Facilitated diffusion of glucose.
 - D) Active transport of sodium ions.
 - E) Movement of lipids across the bilayer.
- B19. A biologist finds two populations of frogs that live in the same area but breed at different times of year. Which form of reproductive isolation is likely preventing gene flow between them?
- A) Geographic isolation
 - B) Behavioral isolation
 - C) Temporal isolation
 - D) Mechanical isolation
 - E) Hybrid inviability
- B20. Which of the following best describes a silent mutation?
- A) A mutation that causes a stop codon to appear early, ending translation prematurely.
 - B) A mutation that changes an amino acid in the protein sequence.
 - C) A mutation that does not change the amino acid specified by the codon.
 - D) A mutation that deletes an entire gene.
 - E) A mutation that shifts the reading frame, altering every codon after it.

C01. How many atoms are there in a nitrate ion?

- A) 1 B) 2 C) 3 D) 4 E) 5

C02. What are the products of the reaction when solid aluminum hydroxide, $\text{Al}(\text{OH})_3$, is reacted with aqueous hydrochloric acid, HCl ?

- A) H_2O , Al , and Cl_2
 B) H_2O and AlCl_3
 C) H^+ , OH^- , and AlCl
 D) H_2 , O_2 , and AlCl_3
 E) H_2O , H_2 , and AlCl_3

C03. The guideline that says atoms and ions tend to be stable when their valence shell contains exactly eight electrons is known as

- A) Kinderhook's Law
 B) Henry the Eighth's Law
 C) Ocho Osos
 D) The Octet Rule
 E) The Rule of Eight

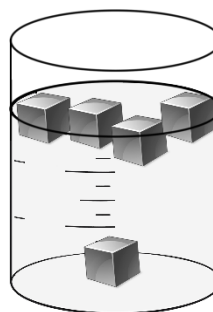
C04. A covalent double bond such as the one found between the carbon atom and each oxygen atom in a carbon dioxide molecule contains

- A) 2 σ bonds
 B) 2 π bonds
 C) 1 σ bond and 1 π bond
 D) 2 σ bonds and 1 π anti-bond
 E) 1 σ anti-bond and 2 π bonds

C05. What is the mass of helium gas in a fully inflated standard-sized 11" party balloon (call it 14.0 L) at 25°C and 1 atm?

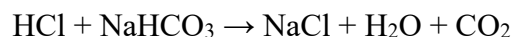
- A) 2.29 g
 B) 2.50 g
 C) 4.00 g
 D) 0.625 g
 E) 0.573 g

C06. The five cubes shown in this diagram are made of copper, lead, tungsten, aluminum, and steel, and they have been dropped into a container of liquid mercury. Which cube is the one at the bottom?



- A) Copper
 B) Lead
 C) Tungsten
 D) Aluminum
 E) Steel

C07. What is the sum of the coefficients when the reaction between hydrochloric acid and sodium bicarbonate is balanced using the smallest whole numbers?

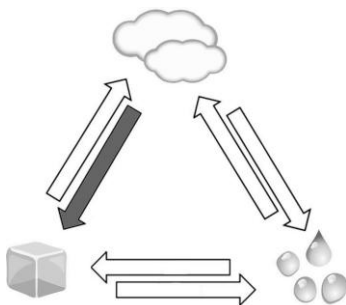


- A) 6 B) 7 C) 8 D) 4 E) 5

C08. If you add 10.0 grams of solid NaCl to a 1 L solution that is saturated in aqueous NaCl , how much solid NaCl will remain undissolved on the bottom of the container? (Ignore any temperature or volume changes due to adding the solid NaCl .)

- A) 10.0 g – none of the added NaCl will dissolve
 B) 0.0 g – the added NaCl will dissolve
 C) less than 10.0 g, but more than 0 g – some of the added NaCl will dissolve, but because the solution is already saturated not all of it will dissolve
 D) more than 10.0 g – the added NaCl will cause some of the dissolved NaCl to precipitate out of the solution.
 E) none of the above answers are correct

C09. What is the name of the phase change indicated by the solid arrow?



- A) Sublimation
- B) Deposition
- C) Solidification
- D) Fusion
- E) Condensation

C10. In a chemical reaction that has reached equilibrium

- A) the reaction has stopped
- B) the reaction continues in a net forward direction, but slowly
- C) the moles of products are equal to the amount of reactants
- D) the concentration of the products is equal to the concentration of the reactants
- E) none of the above answers are correct

C11. For the dissociation of a generic weak acid HA, the ionization constant K_a is equal to

- A) $[HA]$
- B) $[H^+][A^-]$
- C) $\frac{[H^+][OH^-]}{[HA]}$
- D) $\frac{[H^+][A^-]}{[HA]}$
- E) $[HA][H^+][OH^-]$

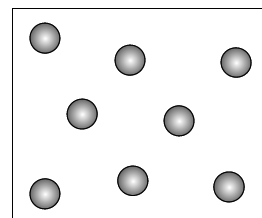
C12. When an atom undergoes oxidation, the atom

- A) gains electrons, leading to a higher oxidation number
- B) loses electrons, leading to a lower oxidation number
- C) gains electrons, leading to a lower oxidation number
- D) loses electrons, leading to a higher oxidation number
- E) becomes an oxygen atom

C13. If you want to determine the instantaneous rate of a chemical reaction at any given point in time, which of these would you use to calculate it?

- A) the equilibrium expression for the reaction
- B) the law of mass action for the equation
- C) the rate law for the reaction
- D) the integrated rate law for the reaction
- E) the Arrhenius equation

C14. In this diagram of a gas sample, each sphere represents one atom. Which of these gases could it be?

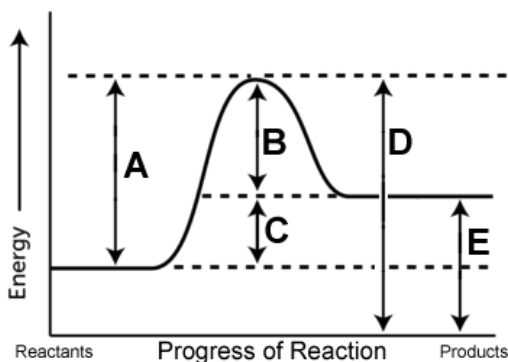


- A) Nitrogen
- B) Argon
- C) Oxygen
- D) Carbon Dioxide
- E) Fluorine

C15. A partial neutralization reaction is carried out by adding 230 mL of 0.025 M NaOH to 120 mL of 0.050 M HCl. What is the pH of the resulting solution?

- A) 1.30
- B) 3.15
- C) 5.65
- D) 7.00
- E) 9.65

- C16. In the potential energy diagram shown below, which arrow corresponds to the activation energy for the forward reaction?



- A) A
B) B
C) C
D) D
E) E
- C17. Given this word equation:
“Ethylene glycol reacts with oxygen to produce carbon dioxide and water.”
What type of reaction is this?
A) Acid-base
B) Synthesis
C) Combustion
D) Double Replacement
E) Single Replacement
- C18. Which of these compounds has the highest percent copper by mass?
A) CuO
B) CuCl₂
C) CuSO₄
D) Cu(NO₃)₂
E) CuS
- C19. What is the solid product formed when aqueous Ba(NO₃)₂ is mixed with aqueous NaBrO₃?
A) barium nitrate
B) barium oxide
C) barium bromide
D) barium bromite
E) barium bromate
- C20. How many coulombs of charge are necessary to reduce 0.200 mol of Ag⁺ ions to solid Ag?
A) 19,297 C
B) 96,485 C
C) 107.87 C
D) 539.35 C
E) 1.20×10^{23} C

P01. According to Malley, in 1895, Wilhelm Röntgen found a new type of invisible radiation which could penetrate solid, opaque objects. What did Röntgen discover?

- A) Infrared radiation
- B) Gamma rays
- C) Beta particles
- D) X-rays
- E) Neutrinos

P02. According to Malley, early in her career, Marie Sklodowska (Curie) studied _____ for the Society for the Encouragement of National Industry.

- A) the magnetic properties of steel
- B) the alloying properties of aluminum
- C) the electrical properties of iridium
- D) the physical properties of tungsten
- E) the radioactive properties of radium

P03. According to Malley, Pierre Curie and his brother Jacques discovered a material that gave off electrical signals when compressed. This is known as the piezo-electric effect. With what material did the Curies discover this effect?

- A) diamonds
- B) garnets
- C) thorium crystals
- D) uranium crystals
- E) quartz crystals

P04. A star that is powered by hydrogen fusion in its core and is about midway through its lifetime is classified as a

- A) regular type star
- B) main sequence star
- C) common sequence star
- D) ordinary star
- E) proto star

P05. For the calculation shown, which answer choice has the correct number of significant figures?

$$X = \frac{11.31 - 6.403}{1.2188}$$

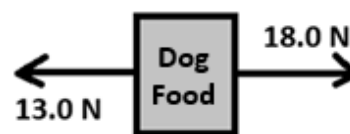
- A) $X = 4$
- B) $X = 4.0$
- C) $X = 4.03$
- D) $X = 4.026$
- E) $X = 4.0261$

P06. You startle a squirrel, which accelerates horizontally from rest at a constant rate of 6.65m/s^2 . How long does it take for the squirrel to reach its maximum speed of 11.4m/s ?

- A) 0.857 sec.
- B) 1.31 sec.
- C) 1.71 sec.
- D) 2.62 sec.
- E) 3.43 sec.

P07. Two horizontal forces act on a box of dog food, as shown. The mass of the box of food is 5.75kg . What is the magnitude of the acceleration of the box of food?

- A) 0.870 m/s^2
- B) 2.26 m/s^2
- C) 3.13 m/s^2
- D) 3.86 m/s^2
- E) 5.39 m/s^2



P08. You use a fixed vertical spring to launch a rock straight up into the air. The spring constant of the spring is 240N/m , and the rock's mass is 42.0g . You compress the spring by 5.50cm , place the rock on it, and release the spring. What is the velocity of the rock at the instant that it launches from the spring?

- A) 2.94 m/s
- B) 4.03 m/s
- C) 5.88 m/s
- D) 16.2 m/s
- E) 31.4 m/s

P09. A meter stick is placed horizontally with a fulcrum set at the 50.0cm mark. A coin with an unknown mass is attached to the meter stick at the 80.0cm mark. A known mass of 130.0g is attached to the meter stick and its position is adjusted to balance the unknown mass. The system balances, and is in perfect equilibrium, when the known mass is located at the 10.7cm mark on the meter stick. What is the mass of the coin?

- A) 17.4 g
- B) 46.4 g
- C) 62.9 g
- D) 170 g
- E) 364 g

P10. You own a pendulum that has a period of oscillation of 1.50sec on Earth. What would be the period of oscillation of this same pendulum if you were to swing it on the Moon? [Note: the acceleration due to gravity on the Moon is 1.62m/s^2]

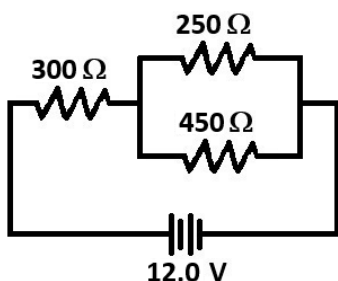
- A) 0.248 sec.
- B) 0.610 sec
- C) 1.50 sec.
- D) 3.70 sec.
- E) 9.07 sec.

P11. A small stone has a mass of 117.0g. When placed in a graduated cylinder containing 60.0ml of water, the stone causes the water level to rise to 96.0ml. What is the density of the stone?

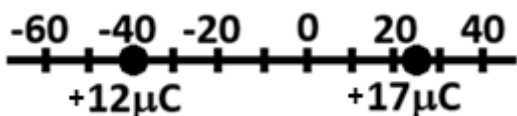
- A) 750 kg/m^3
- B) 1220 kg/m^3
- C) 1950 kg/m^3
- D) 2600 kg/m^3
- E) 3250 kg/m^3

P12. Determine the current flowing in the 450Ω resistor in the circuit shown.

- A) 9.3 mA
- B) 14 mA
- C) 17 mA
- D) 26 mA
- E) 40 mA



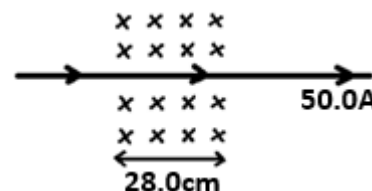
P13. Two charges are placed as shown. A $+12.0\mu\text{C}$ charge is located at $x = -40.0\text{cm}$, and a $+17.0\mu\text{C}$ charge is located at $x = +25.0\text{cm}$. What is the magnitude of the force on the $+17.0\mu\text{C}$ charge due to the $+12.0\mu\text{C}$ charge?



- A) 1.83 N
- B) 2.82 N
- C) 4.34 N
- D) 7.34 N
- E) 11.5 N

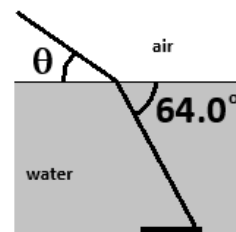
P14. A long straight wire carries a current of 50.0A, as shown. The wire passes through a region with a 400.0 Gauss magnetic field, oriented perpendicular to the wire. The region of magnetic field is 28.0cm wide. What is the net magnetic force acting on the current carrying wire?

- A) 0.56 N
- B) 2.0 N
- C) 7.1 N
- D) 14 N
- E) 35 N



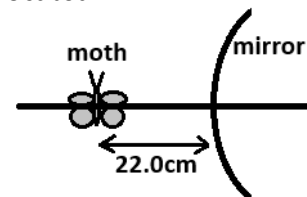
P15. Sadly, your cell phone has been dropped into a lake. A beam of light from your phone shines as illustrated. At what angle, θ , relative to the surface of the water, does the beam of light travel? The index of refraction of the lake water is 1.33.

- A) 35.7°
- B) 42.5°
- C) 47.5°
- D) 54.3°
- E) 64.0°



P16. A convex mirror is placed in a convenience store to give the clerk a good view of the entire store. The magnitude of the radius of curvature of the mirror is 72.0cm. A moth flutters in front of the mirror, at a distance of 22.0cm from the mirror's surface, as shown. How far from the mirror's surface is the virtual image of the moth located?

- A) 9.90 cm
- B) 13.7 cm
- C) 16.9 cm
- D) 31.7 cm
- E) 56.6 cm



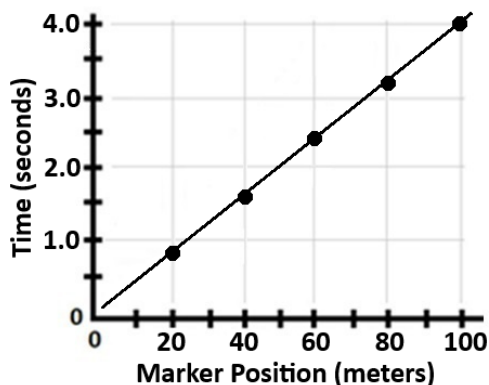
P17. An electron is confined in a quantum dot that has a width of 25.0nm. If we use the size of the quantum dot as the uncertainty in position of the electron, then what is the approximate uncertainty in the velocity of the electron?

- A) $2 \times 10^2\text{ m/s}$
- B) $2 \times 10^3\text{ m/s}$
- C) $2 \times 10^4\text{ m/s}$
- D) $2 \times 10^5\text{ m/s}$
- E) $2 \times 10^6\text{ m/s}$

P18. A meson is discovered that has a mass of about $2000 \text{ MeV}/c^2$, an average lifetime of 5.0×10^{-13} seconds, a charge number of -1 , strangeness of -1 , and charm of -1 . What is the quark structure of this meson?

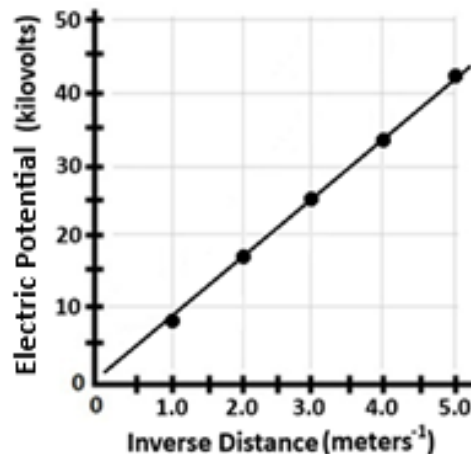
- A) usc
- B) dsc
- C) $\bar{u}s$
- D) $\bar{c}d$
- E) $\bar{c}s$
- F) $c\bar{s}$

P19. The following plot shows the times at which a galloping horse passes fixed position markers along a race track. What is the speed at which the horse is galloping?



- A) 85 m/s
- B) 70 m/s
- C) 55 m/s
- D) 40 m/s
- E) 25 m/s

P20. The plot shown below gives the electric potential (V) measured near a point charge, Q . The potential is plotted against the inverse of the distance from the charge ($1/r$). From these data, determine the approximate magnitude of the charge Q .



- A) 600 nC
- B) 800 nC
- C) 950 nC
- D) 1100 nC
- E) 1300 nC

1A 1	Chemistry																8A 18
1 H 1.01	2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp}	$= 0^{\circ}\text{C}$
T_{bp}	$= 100^{\circ}\text{C}$
c_{ice}	$= 2.09 \text{ J/g}\cdot\text{K}$
c_{water}	$= 4.184 \text{ J/g}\cdot\text{K}$
c_{steam}	$= 2.03 \text{ J/g}\cdot\text{K}$
ΔH_{fus}	$= 334 \text{ J/g}$
ΔH_{vap}	$= 2260 \text{ J/g}$
K_{f}	$= 1.86 ^{\circ}\text{C}/m$
K_{b}	$= 0.512 ^{\circ}\text{C}/m$

Constants

R	$= 0.08206 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$
R	$= 8.314 \text{ J}/\text{mol}\cdot\text{K}$
R	$= 62.36 \text{ L}\cdot\text{torr}/\text{mol}\cdot\text{K}$
e	$= 1.602 \times 10^{-19} \text{ C}$
N_{A}	$= 6.022 \times 10^{23} \text{ mol}^{-1}$
k	$= 1.38 \times 10^{-23} \text{ J/K}$
h	$= 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
c	$= 3.00 \times 10^8 \text{ m/s}$
R_{H}	$= 2.178 \times 10^{-18} \text{ J}$
m_{e}	$= 9.11 \times 10^{-31} \text{ kg}$
$\mathcal{F}$	$= 96,485 \text{ C/mol e}^{-}$
1 amp	$= 1 \text{ C/sec}$
1 mol e^{-}	$= 96,485 \text{ C}$

Densities of Selected Metals (g/mL)

Copper	8.96
Aluminum	2.71
Mercury	13.6
Tungsten	19.3
Osmium	22.6
Steel	7.85
Lead	11.3
Platinum	21.5

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Physics Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles → meters inches → centimeters	1.0 mile = 1609 meters 1.00 inch = 2.54 centimeters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3
Magnetic Field Conversion	Gauss → Tesla	$10^4 \text{ Gauss} = 1.00 \text{ Tesla}$

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2026 INVITATIONAL A**

Biology

B01. E
B02. D
B03. C
B04. B
B05. D
B06. C
B07. A
B08. A
B09. D
B10. E
B11. C
B12. A
B13. D
B14. B
B15. E
B16. B
B17. B
B18. D
B19. C
B20. C

Chemistry

C01. D
C02. B
C03. D
C04. C
C05. A
C06. C
C07. E
C08. A
C09. B
C10. E
C11. D
C12. D
C13. C
C14. B
C15. B
C16. A
C17. C
C18. A
C19. E
C20. A

Physics

P01. D
P02. A
P03. E
P04. B
P05. C
P06. C
P07. A
P08. B
P09. D
P10. D
P11. E
P12. A
P13. C
P14. A
P15. D
P16. B
P17. B
P18. E
P19. E
P20. C

CHEMISTRY SOLUTIONS – UIL INVITATIONAL A 2026

- C01. (D) The nitrate ion is NO_3^- , so there are four atoms.
- C02. (B) This is an acid-base reaction: $\text{Al}(\text{OH})_3(s) + 3 \text{HCl}(aq) \rightarrow \text{AlCl}_3(aq) + 3 \text{H}_2\text{O}(\ell)$
- C03. (D)
- C04. (C) A single bond and the first bond of a multiple bond are sigma bonds. The second bond in a double bond and the second and third bond in a triple bond are pi bonds.
- C05. (A) $n = PV/RT = (1)(14.0)/(0.08206)(298) = 0.5725 \text{ mol}$.
 $0.5725 \text{ mol} \times 4.00 \text{ g/mol} = 2.29 \text{ g}$
- C06. (C) Tungsten is the only one of the five metals whose density is greater than mercury's.
- C07. (E) $\text{HCl}(aq) + \text{NaHCO}_3(aq) \rightarrow \text{NaCl}(aq) + \text{H}_2\text{O}(\ell) + \text{CO}_2(g)$
- C08. (A)
- C09. (B)
- C10. (E) At equilibrium the forward and reverse reactions continue, but at equal rates, so there is no net change in the amounts present. The actual ratio of reactants to products at equilibrium will vary depending on the reaction.
- C11. (D) For any aqueous equilibrium, the equilibrium constant is equal to the concentrations of the products divided by the concentrations of the reactants (leaving out anything that is solid or liquid). Since the dissociation equation for a weak acid is $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$, the products are H^+ and A^- and the reactant is HA .
- C12. (D)
- C13. (C) The rate law expresses the rate of the reaction (typically units like M/sec) based on the concentrations of the reactants at any point in the reaction. The general form of the rate law for a reaction such as $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$ is $\text{rate} = [\text{A}]^x[\text{B}]^y$, where x and y are usually 0, 1, or 2.
- C14. (B) Argon is the only monatomic gas among the answer choices. The other gases are all diatomic.
- C15. (B) Moles $\text{OH}^- = 0.230 \times 0.025 = 0.00575$. moles $\text{H}^+ = 0.120 \times 0.050 = 0.00600 \text{ mol}$.
Net 0.00025 mol H^+ in a total volume of $0.350 \text{ L} = 0.0007143 \text{ M H}^+$.
 $\text{pH} = -\log(0.0007143) = 3.15$
- C16. (A)
- C17. (C) A molecule reacting with O_2 to produce CO_2 and H_2O is a combustion reaction.
- C18. (A) The compound with the highest percent Cu by mass is the one that has the lowest mass of other stuff in it. SO_4 and NO_3 obviously have more mass than O, and so do Cl and S, so CuO has the highest percent copper by mass.
- C19. (E) Barium bromide is BaBr_2 and barium bromite is $\text{Ba}(\text{BrO}_2)_2$.
- C20. (A) The Faraday constant $\mathcal{F}$ (see the data sheet) is the total charge on one mole of electrons and is equal to $96,485 \text{ C/mol}$. To reduce 0.200 mol of Ag^+ ions to solid Ag you need 0.200 mol of electrons. $96,485 \text{ C/mol} \times 0.200 \text{ mol} = 19,297 \text{ C}$.

PHYSICS SOLUTIONS – UIL INVITATIONAL A 2026

- P01. (D) page 10: "...just at the end of 1895.... Wilhelm Röntgen, professor of physics at Würzburg University in Bavaria, had found a new type of invisible radiation which could penetrate solid, opaque objects! Röntgen's rays came from vacuum tubes designed to study cathode rays. He called this radiation 'X' because nothing was known about it."
- P02. (A) page 21: "Marie then began a study of the magnetic properties of different kinds of steel for the Society for the Encouragement of National Industry."
- P03. (E) page 24: "Some years earlier Pierre Curie and his brother Jacques had found that quartz crystals gave off electrical signals when they were compressed. Pierre used this property (known as the *piezo-electric effect*) to devise an unusually sensitive electroscope...."
- P04. (B) When plotted on a Hertzsprung-Russell diagram (a plot of luminosity versus temperature) stars show several distinct groupings. Groups of giant and supergiant stars, as well as protostars and white dwarves can be identified. Most stars, however, fall along a diagonal line from the upper left to the lower right of the diagram. These are the main sequence stars. The main sequence represents where a star will spend most of its lifetime, and the time during which it is powered by hydrogen fusion in its core.
- P05. (C) Correct order of operation requires that we do the subtraction in the numerator first. For subtraction (and addition) it is the decimal place that matters: the answer will only be significant to the same place as the input number with the least decimal places. The number 11.31 is significant to the hundredths place, while the number 6.403 is significant to the thousandths place. The least of these is the hundredths place, so the result of the subtraction will be significant only to the hundredths place:
 $11.31 - 6.403 = 4.90[7]$. The "7" in this case is not significant, but we keep it because we will not round until the entire calculation is finished.
For the division, the result will have the same total number of significant figures as the input number with the least amount of significant figures. The numerator 4.90[7] has three significant figures. The denominator 1.2188 has five significant figures. So, the result will have the least of these – three significant figures. Thus, $X = \frac{4.90[7]}{1.2188} = 4.02[6091] \approx 4.03$.
- P06. (C) The kinematic equation to use is $v_f = v_i + at$. The squirrel starts from rest, so $v_i = 0$. Putting in the final velocity and the acceleration gives: $11.4 = 0 + (6.65)t \rightarrow t = 1.71s$.
- P07. (A) The diagram given also represents the horizontal component of the free-body force diagram, so we have what we need to solve the problem. The two forces act in opposite directions, so the net horizontal force acting on the box is the difference of the two forces: $\sum F_x = 18.0 - 13.0 = 5.0 N$ (to the right, in this case). Now we use Newton's second law to find the acceleration: $\sum F_x = ma \rightarrow 5.0N = (5.75kg)a \rightarrow a = 0.870m/s^2$. Note: there is no motion in the vertical direction, so we do not need to consider any vertical forces.
- P08. (B) Initially, we have elastic potential energy stored in the compressed spring. Once released, this elastic potential energy converts into the kinetic energy of the rock. There is also a little gravitational potential energy resulting from the lifting of the rock as the spring uncompresses. Thus, we have:
 $EPE_i = KE_f + GPE_f \rightarrow \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + mgx \rightarrow$
 $(240)(0.055)^2 = (0.042)v_f^2 + (0.012)(9.80)(0.055)$. This gives a launch velocity of the rock of
 $0.726 = 0.042v_f^2 + 0.0065 \rightarrow v_f = 4.03m/s$.

- P09. (D) We only need to consider rotational equilibrium to solve the problem. Let's start with the torques acting on the meter stick. Torques are produced by both the known and unknown masses. The normal force at the fulcrum does not provide a torque since it acts at the axis of rotation and has a torque arm of zero. The known mass produces a torque oriented in the counterclockwise direction, while the unknown mass produces a clockwise-oriented torque. Thus, we have the rotational equilibrium equation:
 $\sum \tau = 0 \rightarrow m_k g r_k \sin \theta_k - m_u g r_u \sin \theta_u = 0$. The gravitational forces are in the vertical direction (downward), and the meter stick is horizontal, so both angles are $\theta_k = \theta_u = 90.0^\circ$. The fulcrum serves as the axis of rotation, so all torque arm distances will be measured from the 50.0cm mark. For the unknown mass, the torque arm is $r_u = 80.0 - 50.0 = 30.0\text{cm}$. For the known mass, the torque arm is $r_k = 50.0 - 10.7 = 39.3\text{cm}$. Putting in the known values gives:
 $(0.130)(9.80)(0.393) \sin(90.0) - m_u(9.80)(0.300) \sin(90.0) = 0$. This leads to $0.501 = 2.94m_u$ and an unknown mass of $m_u = 0.170\text{kg} = 170\text{g}$.
- P10. (D) The equation for the period of a pendulum is $T = 2\pi \sqrt{\frac{L}{g}}$. We do not know the length of the pendulum, but we can find that using the information given. On Earth: $T = 1.50\text{s} = 2\pi \sqrt{\frac{L}{(9.80)}} \rightarrow$
 $L = (9.80) \left(\frac{1.50}{2\pi} \right)^2 = 0.5585\text{m}$. Now, on the Moon, the length of the pendulum is the same, but we have a different gravitational acceleration, so we get a period of $T = 2\pi \sqrt{\frac{L}{g_m}} = 2\pi \sqrt{\frac{0.5585}{1.62}} = 3.70\text{s}$.
- P11. (E) We have the mass of the stone, so we only need to determine its volume. The volume of the stone equals the amount that the water level rose when the stone was placed in the graduated cylinder. That is $V_{\text{stone}} = 96.0\text{ml} - 60.0\text{ml} = 36.0\text{ml}$. The density is then
 $\rho = \frac{m_{\text{stone}}}{V_{\text{stone}}} = \frac{117.0\text{g}}{36.0\text{ml}} = \frac{0.1170\text{kg}}{0.0000360\text{m}^3} = 3250\text{kg/m}^3$.
- P12. (A) First, we combine the two resistors that are in parallel: $\frac{1}{R_p} = \frac{1}{250} + \frac{1}{450} \rightarrow R_p = 160.7\Omega$. The parallel group is in series with the remaining resistor. Combining these gives the total resistance of the circuit: $R_T = 300 + 160.7 = 460.7\Omega$. Now we can find the current produced by the power source:
 $V_T = I_T R_T \rightarrow 12.0 = I_T(460.7) \rightarrow I_T = 0.0260\text{A}$. Because they are in series, this same current flows through both the 300Ω resistor and the 160.7Ω parallel group ($I_{300} = I_p = I_T = 0.260\text{A}$). Focusing on the parallel group, we find the voltage across the parallel group: $V_p = I_p R_p = (0.260)(160.7) = 4.19\text{V}$. Because the resistors are in parallel, this is the same voltage across the 250Ω and 450Ω resistors ($V_{250} = V_{450} = V_p = 4.19\text{V}$). Now we can find the current flowing through the 450Ω resistor:
 $V_{450} = I_{450} R_{450} \rightarrow 4.19 = I_{450}(450) \rightarrow I_{450} = 0.0093\text{A} = 9.3\text{mA}$.
- P13. (C) The force of one charge on another is given by Coulomb's Law. Since we aren't concerned about direction, we can use the equation without worrying about the vector: $|F| = \frac{kQ_1Q_2}{r^2}$. The distance between the charges is $r = 25\text{cm} - (-40\text{cm}) = 65\text{cm} = 0.65\text{m}$. So, the force on the $17.0\mu\text{C}$ charge due to the $12.0\mu\text{C}$ charge is $|F| = \frac{(8.99 \times 10^9)(12 \times 10^{-6})(17 \times 10^{-6})}{(0.65)^2} = 4.34\text{N}$.
- P14. (A) The force on a current-carrying wire due to a magnetic field is given by $F = ILB \sin \theta$. Since the magnetic field is oriented perpendicular to the wire, we have $\theta = 90.0^\circ$. We also convert the magnetic field strength into Teslas: $B = 400.0\text{G} = 0.04000\text{T}$. Plugging it all in, we obtain a force of
 $F = ILB \sin \theta = (50.0)(0.280)(0.04000) \sin(90.0) = 0.56\text{N}$.

- P15. (D) The angle given is relative to the surface of the water, but we need to find the angle relative to the normal line to the surface. That is, the incident angle is $\theta_i = 90.0 - 64.0 = 26.0^\circ$. It is this angle which we use with Snell's Law. Since the light beam starts in the water, the initial index of refraction is 1.33, and the final index of refraction is 1.00 (air). Using Snell's Law, we get $n_i \sin \theta_i = n_r \sin \theta_r \rightarrow (1.33) \sin(26.0) = (1.00) \sin \theta_r \rightarrow 0.583 = \sin \theta_r$. So, the refracted angle is $\theta_r = 35.7^\circ$. This is the angle relative to the normal line to the surface. Thus, the angle of the beam in air relative to the surface of the water is $\theta = 90.0 - 35.7 = 54.3^\circ$.
- P16. (B) The mirror is convex, so the focal length is negative. We calculate a focal length for the mirror of $f = (-)\frac{R}{2} = -\frac{72.0}{2} = -36.0\text{cm}$. The object (the moth) is located at $p = 22.0\text{cm}$. Thus, we can find the image location by using $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \rightarrow \frac{1}{22.0} + \frac{1}{q} = \frac{1}{-36.0} \rightarrow q = -13.7\text{cm}$. Since the image location is negative, this confirms that the image is virtual (as stated), and the magnitude of the distance of the virtual image from the mirror's surface is $|q| = 13.7\text{cm}$.
- P17. (B) The uncertainty principle for position and momentum is given by $\Delta x \Delta p \geq \frac{\hbar}{2}$. As usual, we will use the equality to estimate the uncertainty. Putting in the given uncertainty in position, and inserting the value for $\hbar$, we obtain: $(25.0 \times 10^{-9})\Delta p = \frac{1.054 \times 10^{-34}}{2} \rightarrow \Delta p = 2.11 \times 10^{-27}\text{kgm/s}$. This is the uncertainty in momentum. The electron is trapped, and this small uncertainty in momentum is non-relativistic, so we can relate the uncertainty in momentum to the uncertainty in velocity using classical equations: $\Delta p = m\Delta v \rightarrow 2.11 \times 10^{-27} = (9.11 \times 10^{-31})\Delta v$. This gives an uncertainty in velocity of $\Delta v = 2.3 \times 10^3 \approx 2 \times 10^3\text{ m/s}$. Note: You may find variations in the equation for the uncertainty principle, but all of them will lead to an order of magnitude of $\Delta v \approx 10^3$.
- P18. (E) Since this particle is a meson, it must consist of a quark and an anti-quark. This eliminates choices A and B (which are baryons). The strangeness must be -1 , so the quark must be an s-quark. This eliminates choices D and F. The charm of -1 tells us that the anti-quark must be an anti-c-quark. Thus, the correct answer is (anti-c)(s), or $\bar{c}s$, which is choice E. This also gives the correct charge number of -1 .
- P19. (E) On a plot of position and time, the velocity is related to the slope of the line. In this case, the slope is the inverse of the velocity, which can be seen from the units of the slope. To begin, let us calculate the slope itself (with units). We first choose two points on the best fit line that are not data points. I shall choose (24m, 1.0s) and (73m, 3.0s). These points give a slope of $\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3.0\text{s} - 1.0\text{s}}{73\text{m} - 24\text{m}} = \frac{2\text{s}}{49\text{m}} = 0.041\text{s/m}$. The velocity is then $v = \frac{1}{\text{slope}} = \frac{1}{0.041\text{s/m}} = 24.5\text{m/s} \approx 25\text{m/s}$.
- P20. (C) Once again, it will be the slope of the line that is important. The electric potential near a point charge is described by the equation $V = \frac{kQ}{r}$. Relating this to our plot shows: $V = (kQ)\left(\frac{1}{r}\right)$. Thus, we see that the slope is equal to kQ . To calculate the slope, we choose two points on the best fit line that are not data points. I shall choose (1.1V, 10m^{-1}) and (3.5V, 30m^{-1}). These points give a slope of $\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{30\text{kV} - 10\text{kV}}{3.5\text{m}^{-1} - 1.1\text{m}^{-1}} = \frac{20000\text{V}}{2.4\text{m}^{-1}} = 8330\text{Vm}$. Setting this equal to the quantity that we derived from the equation gives: $kQ = 8330\text{Vm} \rightarrow \left(8.99 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}\right)Q = 8330\text{Vm}$. This gives $Q = 9.3 \times 10^{-7} \frac{\text{C}^2\text{V}}{\text{Nm}} = 9.3 \times 10^{-7}\text{ C} = 930\text{nC} \approx 950\text{nC}$.